Labor Market Segmentation and the Distribution of Income: New Evidence from Internal Census Bureau Data

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Abstract

In this paper, we present new findings that validate earlier literature on the apparent segmentation of the US earnings distribution. Previous contributions posited that the observed distribution of earnings combined two or three distinct signals and was thus appropriately modeled as a finite mixture of distributions. Furthermore, each component in the mixture appeared to have distinct distributional features hinting at qualitatively distinct generating mechanisms behind each component, providing strong evidence for some form of labor market segmentation. This paper presents new findings that support these earlier conclusions using internal CPS ASEC data spanning a much longer study period from 1974 to 2016. The restricted-access internal data is not subject to the same level of top-coding as the public-use data that earlier contributions to the literature were based on. The evolution of the mixture components provides new insights about changes in the earnings distribution including earnings inequality. In addition, we correlate component membership with worker type to provide a tacit link to various theoretical explanations for labor market segmentation, while solving the problem of assigning observations to labor market segments a priori.

Keyword*:* Inequality, Income Distribution, Mixture Model

JEL Classification: C16, D32, J01

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Introduction

Schneider (2013) proposed that a finite mixture distributional model may be used to reconcile economists traditional intuitions about the shape of the distribution of income with physicists' more recent findings that incomes appear roughly exponentially distributed up to relatively high levels. Using CPS data, the author showed that based on a variety of fit criteria a finite mixture model consisting of an exponential component and a log-normal component fits the data better than either the exponential or the log-normal distributions alone, and better than most alternatives. Schneider (2015) critically revisits the physics literature on the distribution of income and verifies the earlier finding using IRS tax data. In fact, the finite mixture model performs on par with the commonly used generalized beta of the 2nd kind (GB2) distribution used by Feng, Burkhauser, and Butler (2006), Burkhauser, Feng, Jenkins, and Larrymore (2008), Jenkins (2009).

In this paper, we show that the results presented in Schneider (2013) are robust by using internal Census data that is top-coded at a much higher reporting threshold than the public-use data used in the original paper. Specifically, we pick up the research question whether the US income distribution can be characterized by a mixture of frequency distributions that has finite components and that is stable with respect to the number and type of components. In addition to using data with a much higher top-code limit, we provide fits over a much longer period and provide stronger evidence for the long-run stability of the functional features of our model.

The changes in the parameter estimates provide novel insights about the evolution of the income distribution from 1975 to the present. Both Schneider (2013) and Schneider (2015) asserted that the components of the mixture model have more intuitive economic interpretations than more complicated distributions like the GB2. In what follows, we provide a cursory overview of the literature, briefly outline the intuition behind the specific two-component mixture model, and discuss a variety of unresolved issues. We then present the

results for the exponential-log normal mixture based on internal Census data for earnings from 1974 to 2016.

Our project adds to a growing recent literature using publicly available IRS and CPS data to estimate finite mixture models with distinctive components, yielding remarkably accurate fits of the observed frequencies (Dragulescu and Yakovenko, 2001, Yakovenko and Silva, 2005, Yakovenko, 2007, Shaikh, Papanikolaou, and Wiener, 2014, Schneider, 2015). These models effectively identify the presence of at least two distinct and formally persistent "signals" in the U.S. distribution of income: a power-law distribution for high-income individuals (Parker and Fenwick, 1983, Fichtenbaum and Shahidi, 1988, Bishop, Chiou, and Fromby, 1994, Champernowne and Cowell, 1998, Yakovenko, 2007, Philip Armour, 2014) and an exponential. Schneider (2013) and Schneider (2015) further suggest the possibility of an additional log-normal component in the mixture. Since all the work in this literature relies on publicly available data, a verification using the restricted-access data available through the US Census Bureau constitutes an important contribution, not least of all because top-coding and binning procedures for the public-use data periodically change over time. While the work cited above takes great care to adjust appropriately, there is no substitute for the richer restricted-use CPS ASEC data covering a longer study period.

Background

Dagum (1977) contrasted two approaches in inquiry into the observed frequency distributions of income, defined by whether those distributions are taken to represent a tractable expression of the processes conditioning individual incomes or simply fit a shape to the observed distribution. Much recent work effectively rejects the proposition that there is a tractable connection between the best fitting functional form of a distribution fit to the data and the underlying generating mechanism. That is, the observed shape of the distribution of income is not held to reveal insights about the workings of labor markets. Current

approaches instead emphasize the identification of the best-fitting general shapes for the observed distributions, paying little attention to what this may imply for economic theory. Explanatory parsimony is downplayed in favor of goodness of fit. The limit of this approach is a non-parametric fitting of the observed distribution, but in application it is more common to use a very flexible parametric distribution like the GB2. A good representation of researchers working in this second paradigm is given by Feng et al. (2006), Burkhauser et al. (2008), Jenkins (2009). While these more flexible models tend to provide a good fit to the empirical data, they provide little in the way of a theoretical explanation of the generative processes and a priori exclude the possibility of multiple distinct labor markets each with their own statistical signature. Before revisiting why this appears to be the preferred approach at present, we want to outline the alternative, which is the focus of this paper, has a long pedigree in the history of economic thought.

The alternative approach seeks to understand well-fitting distributions as expressions of the underlying market processes generating income. Work following this approach seeks to establish analytical linkages between the distribution(s) it identifies and plausible economic behavior or functioning. Parsimony and the ability to inform economic theorization are the central principles of analysis. This approach mirrors the methodological approach physicists take to studying distributional data. Its history parallels the history of advances in Statistical Mechanics, where the Principle of Maximum Entropy allowed formal consideration of the relationship between the micro-level "laws of motion" governing the interactions between individual particles and the macroscopic properties of large collections of particles, including the distributions of certain quantities across particles.

The earliest contributions in this tradition date back to the turn of the last century. Vilfredo Pareto devoted considerable effort to the construction of income and wealth frequency distributions for several countries, leading to his discovery at the very top of income ranges of the kind of "fat tail" distributions that today bear his name. Pareto postulated this com-

mon distribution constituted a fundamental "social law" in capitalist economies. This work was furthered by Gibrat (1931), who considered that the growth of enterprises followed geometric processes, which he associated with log-normal distributions for their size. Analogous multiplicative "laws of motion" were soon put forward for income-generating processes and linked to specific predictions concerning individual distributions of income. Kalecki (1945) pointed out that such a process (e.g. if experience and education increase worker skill in a multiplicative, rather than additive, fashion) could lead both to log-normal and to power-law distributions (see also Sutton (1997)).

Problems for this approach arose by the middle of the 20th century, when Lydall (1959) showed that the observed distribution of income did not appear to be log-normal. More recent contributions grounded on this methodological approach have consistently uncovered evidence for power-law behavior among high incomes and rejected the proposition that incomes were log-normally distributed, but they failed to settle on a convincing fit for the bulk of the distribution (Dagum, 1977, Bordley, McDonald, and Mantrala, 1997, Champernowne and Cowell, 1998). The shift towards simple shape-fitting was at least partially motivated by the perceived failure to conclusively identify how the bulk of incomes are distributed. One casual justification for shape-fitting was a collective resignation that the observed distribution of income pooled the outcomes from many different process. Given a sufficiently large number of diverse processes, any distributional shape could arise and identification of a best-fitting distribution would be a futile endeavor.

Concurrently, economists developed ideas about labor market segmentation that on the one hand would indeed suggest that the observed distribution of income pools observations from different processes, but also that the number of processes is finite and limited in number. Furthermore, one can extrapolate that each process has distinct features, so that under the assumption that markets are entropy maximizing process (Foley, 1994), each labor market segment should be assumed to produce a distinct stable distributional signature. However,

this line of reasoning was not pursued until Schneider (2013).

Partially, this may have been because one version of labor market segmentation, the "dual labor market" theory of Reich, Gordon, and Edwards (1973), pursued a more narrative approach for characterizing the qualitative differences between labor market processes. Other explanations for segmentation provided formal models, but none of them explored the implications for the stable shape of the observed distribution (see Weitzman, 1989, for example). This reflects in part a disciplinary limit in how economists tend to think about equilibrium. Economists instinct is to think of prices as uniquely determined and thus any distribution of prices to only reflect some underlying distribution of determinants (e.g. innate ability, human capital, etc.). It is exactly in this regard that (Foley, 1994) offers the novel deviation by showing what happens when the restriction that agents only make maximizing exchanges is relaxed to allow all Pareto-improving exchanges. The result is that a statistical equilibrium distribution of prices emerges that is the result of an entropy maximizing process. The empirical implication is that if we can identify the distribution consistent with this statistical equilibrium, it can help us characterize the specific features of the underlying market process as the constraints in an entropy maximization program (MaxEnt).

This approach, however, also immediately implies some intellectual limitations. The reverse implication of the this approach is that *any* process that is consistent with the identified constraints in the MaxEnt program is a possible explanation for the data. That is, this approach allows us to demarcate between theories that have distinctly different distributional implications, but not between theories that have the same distributional implications. For instance, after convincing themselves that incomes appeared exponentially distributed, physicists have posited a conservation law for the total wage-bill while economists (who reject this notion) might point to the mean-preserving growth or a "social scaling" mechanism (dos Santos, 2017). Since all of these are consistent with the exponential arising as an entropy maximizing distribution in statistical equilibrium, the observation of the exponential cannot

help differentiate between these hypotheses. Only additional information that leads us to favor the premises behind one position over the others could be used to demarcate between them.

It may seem like a basic point, but it has direct bearing on the results we are presenting in this paper and how we interpret them. Specifically, we show empirical evidence for the number and plausible type of distributional components, but do not endeavor to discriminate which plausible explanation for labor market segmentation may be ruled out by these findings. More work needs to be done to answer this broader question with any satisfaction. Suffice it to say for the purposes of this paper that there several explanations for segmentation have been put forth and we remain agnostic about which one we favor. More detailed investigation may reveal that not all can be reconciled equally well with our findings, but confounding the issue is that they may also not all be mutually exclusive.

For concreteness, the main competing theories we have in mind include the narrative proposed by Reich et al. (1973) that describes a primary labor market bifurcated into subordinate and independent jobs and a secondary labor market comprised of unstable low wage jobs. Weitzman (1989) proposed a formal model based on wage stickyness and uncertainty about labor demand that he suggests is consistent with Reich et al. (1973).[1] The thrust of Weitzman (1989)'s model is that two strategies employer strategies evolve and may exist in what he calls *sticky wage stochastic competitive equilibrium* (p. 127): some employers offer higher wages to maintain a stable work force, which works because they also commit to higher per employee capital intensity in production, while other employers insure against economic uncertainty through a low-wage / low-capital intensity strategy in order to "thrust the burden of unstable business on a casual work force." (p. 123) Perhaps most importantly, Weitzman (1989) argues that the emergent pattern of segmentation is likely to be largely in-

[1]Weitzman (1989) concedes that his model is sufficiently abstract to be consistent with a number of different explanations.

dependent of any initial skill distribution, because any arbitrary sorting into one or the other kind of job can result in an employee being labeled in a way that proves self-reenforcing.

In particular, this leaves open a mechanism for characteristics that are unrelated to actual productivity differences to become determinant of a worker's employment trajectory. Economic stratification (see Jr., Hamilton, and Stewart, 2015), for example, seems like an easily reconcilable feature of the kind of segmentation described by Weitzman (1989), where implicit bias, previous wages, differential educational opportunities, etc. all serve as the kind of self-fulfilling labeling system that sorts female and minority workers into lower-wage, precarious employment. A somewhat different argument for a similarly racialized segmentation is given by Temin (2017), who suggests that the US is regressing towards a dual economy as characterized by the Lewis model originally proposed to explain disparate sectors in developing countries. There are clear differences in this conception from the one offered by Weitzman (1989)'s model,[2] though it is unclear to us whether these differences might not be reconcilable.

Dickens and Lang (1993) make the point that any of the diverse appeals to segmentation are a counter to standard human capital theory as an explanation for wage dispersion. For instance, they differ from search-theoretic explanations for wage dispersion, like Montgomery (1991), that tend to either assume or conclude that there is a strict correlation (if not outright equality) between individual worker's pay and they productivity, taken as exogenously determined. This still applies when one reads the authoritative Acemoglu and Autor (2011), who posit wage dispersion in a perfect labor market as a consequence of skill differentials. The key point for the present work is that we look for a statistical signature of segmentation that goes beyond simply the level of wage dispersion. Our interpretation of

[2]Not least, the Lewis model posits that the marginal product of labor in the less developed sector is zero (or approximately zero), whereas the individual worker's marginal product is actually irrelevant in Weitzman (1989).

labor market segmentation is that it implies a finite set of qualitatively different signals in the overall observed distribution of income from labor, which is a more concretely testable claim than simply offering different explanations for an observed growing dispersion between the incomes of different workers.

It appears, however, that this was not fully recognized by the advocates for labor market segmentation either. Osterman (1975) recognizes the identification problem resulting form sorting worker-observations *a priori* and Dickens and Lang (1993) directly address the sentiment that labor market segmentation is an untestable proposition. In fact, Schneider (2013) attributes the recent stagnation of this line of work to the failure to consider finite mixture models, which offer a systematic approach for testing the evidence in favor of segmentation without the pre-sorting problem discussed by Osterman (1975). The core idea is that since different labor market segments operate in a qualitatively different ways, thus generating different stationary distributions, comparing the fit of different specifications of a finite mixture model[3] can reveal both the number and type of segments evident in the data. This paper, however, does not present the comprehensive investigation with respect to number and type of segments supported by the data, but more modestly looks at the robustness of the results presented by Schneider (2013) using a richer dataset over a longer period.

Recent work by Lubrano and Ndoye (2016), Anderson, Farcomeni, Pittau, and Zelli (2016), Anderson, Pittau, Zelli, and Thomas (2018) has applied finite mixtures models to labor market data. While this research has clearly made important contributions to the underlying statistical methodology of finite mixture models it has remained pragmatic about the particular distributional form used, often employing the log normal distribution due to its convenient statistical properties. The ambition of this paper is apply the same statistical methodology of finite mixture models while also giving serious consideration to the distinct

[3]What amounts to a spectral analysis of the earnings distribution.

characteristics of each component. This approach leads us to consider the statistical dissimilarity of each labor market segment. Our results thus extend the work of Lubrano and Ndoye (2016), Anderson et al. (2016, 2018) to the use finite mixture models with diverse components.

Specific Model

Both the narrative theory of Reich et al. (1973) or the model by Weitzman (1989) posit a labor market segment that matches workers to positions that fundamentally do not rely on the individual attributes of a given worker. These may be positions described as "low-skill" (or better, highly routinized) where workers filling them can be treated as homogeneous not because they are, but because their attributes and skills matter very little to getting the job done, thus a relatively constant wage should prevail. As Weitzman (1989) makes clear, the key feature of this segment is that workers absorb all uncertainty and thus pay fluctuations from one period to the next. As we will argue, this combination of relatively constant wage and high earnings churn as the result of precarious employment (or at least hours) is quite consistent with an exponential distribution of earnings. By contrast, another labor market (itself perhaps bifurcated) does match employers and employees based on the individual worker's work history. Here, the multiplicative combination of past experience (and pay) should give rise to either a log-normal or power-law distribution of wages.

We further posit that looking at earnings, we are necessarily conflating the wage and the number of hours worked. However, workers in the latter segment are much more likely to hold full-time positions with an expectation of 40 hours of work per week. Variations in hours is likely to be much less than variation in wages and thus the distribution of incomes should be expected to largely reflect the distribution of wages. On the other hand, workers who fill positions that simply require a body are more likely to occupy jobs with less than full-time hours and have highly variable hours across the year. (Seasonal hires in retail,

services, or agriculture for example.) For this segment, variation in incomes is more likely come from the hours worked than the wages paid. We speculate that it is this latter labor market segment that is captured by the apparent exponential distribution of low to moderate incomes discovered by physicists in the early 2000s (see Dragulescu and Yakovenko, 2001, Yakovenko and Silva, 2005, Yakovenko, 2007). Rather than a conservation law of money (as the physicists in question postulated), we posit that the exponential reflects a relatively constant overall demand for basic labor with a lot of dynamic churn across sectors in a given year as to where exactly basic labor is required.

A simple preliminary test to see if this line of reasoning might prove salient is to see if the specific components in our mixture model are correlated with the appropriate worker type (full-time or part-time). If it is indeed the case that part-time workers dominate the exponential component of the mixture, it also provides a link to the framing provided the literature on stratification. That literature has shown that *who* performs basic labor in the economy – defined by low wages, no benefits, variable hours below full-time – is highly gendered and racialized. Contrary to the popular belief that part-time jobs paying the minimum wage are filled predominantly by high schoolers getting their first work experience, they are much more likely to employ men and women of color past high-school age. If the correlation between worker type holds, it will warrant further investigation whether the expected correlation with worker race and gender also holds, and this will reveal exactly who occupies each of the labor market segments that we identify. It also gives credence to the role of "labeling" mechanisms postulated by Weitzman (1989).

While these considerations present valuable directions for future research, they are beyond the scope of the present paper, which simply shows how an exponential-log normal mixture fits the observed distribution of income, that this fits is stable across a relatively long period, and then tests the correlation between worker type and mixture component to show that indeed part-time workers dominate the exponential component of the mixture.

Methodology

We follow the innovations by Borzadaran and Behdani (2009), Scharfenaker and Semieniuk (2016), Scharfenaker and Foley (2017), Shaikh et al. (2014), dos Santos (2017) that draw on Information Theory and Statistical Mechanics to develop economically meaningful accounts of these inferred distributions. In relation to individual income, the observation of frequencies that are consistently well-described by a specific functional distribution allows the use of the Principle of Maximum Entropy to infer the aggregate, mathematical forms taken by the processes shaping those distributions. Those will be given by the aggregate constraints defining the statistical ensemble over which the observed distributions are entropy maximizing, as argued by E. T. Jaynes. In many instances, it is possible to advance plausible economic processes capable of generating such constraints. For example, exponential distributions may provide evidence that wage incomes embody the competitive "social scaling" of unobservable measures of physical productivity and bargaining power over money wages, as argued in dos Santos (2017); log-normal distributions may in turn reflect a stochastic evolution of income that is on average proportional to previous income. It is thus likely that distinct "signals" in the observed distribution of income reveal the existence of formally distinct modes of income appropriation in line with the theory of labor market segmentation.

A nuance to this approach is to recognize that while it may be the case that the underlying market processes are entropy maximizing (as argued by Foley, 1994, for instance), the Principle of Maximum Entropy put forth by E. T. Jaynes suggests that the solution to a program that maximizes Shannon's entropy is relevant even if the process is not strictly entropy maximizing. Even in this case, the MaxEnt solution presents the most probably description – barring the incorporation of explicit information of the actual process. Put differently, the MaxEnt solution imposes the least restrictive prior when the underlying when it is not known whether the underlying process is entropy maximizing or not. The twist to this underlying

the present investigation is that we assume each mixture component (corresponding to a distinct labor market segment) is best modeled as a maximum entropy distribution, while we do not make any assumptions whether the number of components is somehow entropy maximizing.

Data

Like Schneider (2013), we use CPS person-level earnings data from the ASEC (March Supplement). The variable WSAL-VAL combines wage and salary income across multiple jobs, but excludes self-employment income as well as non-labor income. Because of availability issues, Schneider (2013) was limited to a series of public-use data from 1996 to 2007. Furthermore, the public-use data was subject to relatively low top-code limits that were accounted for in the estimation by using the top-code limit as a censoring limit and using the top-code flag as a count of censored observations.

The analysis in this paper uses restricted-access CPS ASEC data gathered from the Federal Statistical Research Data Center (RDC). With this restricted-access data we can examine the WSAL-VAL variable from 1974 to 2016 which is also subject to a much higher (less restrictive) top-code limit. We still treat top-coded observations as censored and make the appropriate adjustment in our maximum likelihood estimations, but our robustness checks indicate that this has only a minor (and not economically meaningful) impact on the actual estimates.

Top-coding is potentially a major limitation when using the publicly available data. The variable used in our study, WSAL-VAL, is the sum of two variables recorded during data collection: ERN-VAL and WS-VAL, or earnings from primary and secondary jobs respectively. According to the CPS documentation, these variables are top-coded before they are added to create WSAL-VAL. If one or the other, or both, earnings components are top-coded, it is difficult to model the uncertainty that is being introduced through the

truncation procedure. Schneider (2013) dealt with this by dealing with all observations that contained a top-coded value as censored and using the top-code as the censoring limit. Since we are using restricted access data, this is not an issue in the work presented in this paper beyond the censoring of the data above the internal recording limit already mentioned. For more information on the topcoding issue, see Larrimore, Burkhauser, Feng, and Zayatz (2008).

There are several procedural changes to the CPS ASEC dataset during the study window,[4] two of which deserve explicit mention. First, data collection changed from paper-and-pencil to computer assisted in 1993. Along with this change, certain recording limits on income variables were adjusted; most relevantly for us, earnings up to $999,999 were now recorded. There were also expected adjustments to implement the 1990 Census population controls in 1992 and further adjustments in 1994 to update the sample design. All together, the changes made in this period appear to produce a notable "jump" in some estimates presented in this paper (similar jumps are seen in Larrimore et al., 2008, Schneider, 2013, 2015, for example). There was also a significant redesign in 2014 of the CPS ASEC, but it does not appear to produce the kind of discontinuity that resulted from the changes in 1993.

Second, there is a large change in sample size in 2002. From 1975 to 2001, the CPS ASEC contained 60,000 to 87,000 raw observations; from 2002 to 2008, over 100,000 observations were recorded before it slowly declined back down to 87,000 by 2017. However, the March supplement weights used to match the CPS ASEC sample to the larger CPS and last Census seem to successfully adjust for variations in sample sizes such that estimates do not seem to fluctuate with sample size in any notable way because the sample weights, MARSUPWT, sufficiently ensure that samples remain representative of the population.[5]

[4] `https://www.census.gov/topics/income-poverty/income/guidance/cps-methodology-changes.html`

[5] We suspect that Larrimore et al. (2008) may understate the impact of topcoding by focusing on the number of observations subject to topcoding, not their population weight. Since the weights are in principle

The publication of cell means for the topcoded observations starting in 1996 and the subsequent series of cell means for 1976 to 2005 produced by Larrimore et al. (2008) permit a back-of-the-envelope estimation of the fatness of the tail. If we assume that the upper tail follows a power-law (which is not reflected in our model), then the cell mean has a straightforward relationship to the topcode limit and the power-law parameter α. Specifically, if x_{TC} is the topcode limit and $\bar{x}_{CM}$ is the cell mean, then:

$$\alpha = \frac{\bar{x}_{CM}}{\bar{x}_{CM} - x_{TC}} \tag{1}$$

This quick method of estimating the power-law coefficient is only available until the 2010. After that, the public-use data contains income values subject to topcoding, but they where randomized across similar observations. From 1975 to 2009 (adjusting for the fact that the 2010 CPS ASEC asks about income earned during the previous year), the estimated α decreased from over 4 to around 2 as seen in figure 1, implying that the tail of the earnings distribution got a lot fatter. The obvious implication is that inequality increased driven by very high incomes, consistent with the trends described by Piketty (2014) among others.

One important point to keep in mind is that the values censored by the internal reporting limits are used in the calculation of the cell means. So for example, a person reporting an income of more than \$100,000 from their primary job would be captured as having earned \$99,999 and the corresponding cell mean would have been calculated using that lower value. The power-law coefficient estimates based on the cell means therefore likely represent a lower-bound estimate to the fatness of the right tail of the distribution of earnings.

In the model we fit to the data that will be described in the next section, we do not allow

an adjustment for under-sampling, a single observation subject to topcoding may have a significant impact on estimates if it is appropriately weighted.

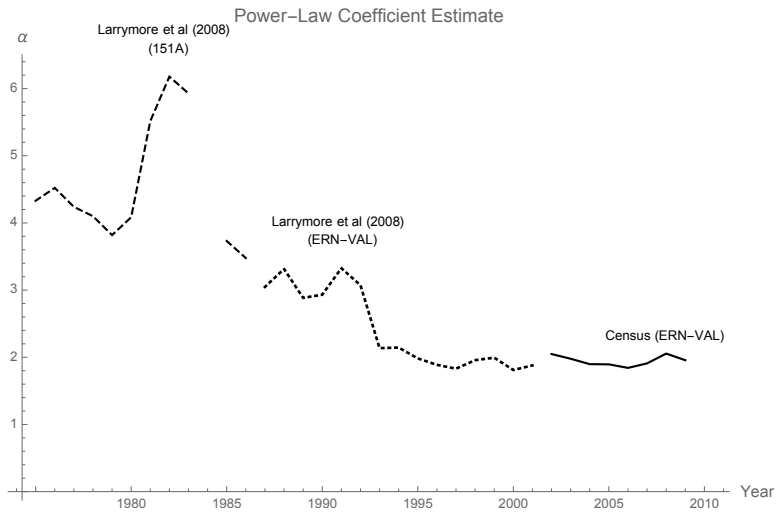


Figure 1: Rough estimate of the "fatness" of the right tail of the primary earnings distribution based on cell means provided by Larrimore et al. (2008) for 1976 to 2002 and part of the CPS ASEC dataset from 2003 to 2010. Estimates are based on the cell mean for male non-Hispanic / black full-time observations.

for a power-law tail for earnings and we make no attempt to test whether high earnings are better fit by a power-law or the log-normal in this paper. Without a rich set of observations from the right tail of the distribution, distinguishing between a power-law and log-normal tail can be statistically difficult. Furthermore, our mixture model may exacerbate the identification problem as the tails of the components are allowed to overlap. That said, we believe there is reason to think that our implicit assumption that the log-normal is sufficient to capture the thickness of the right tail of the earnings distribution may be wrong and that we therefore systematically understate inequality driven by top incomes through our estimates. Correcting for this is part of our continuing work.

Model

The basic idea of using a finite mixture model (FMM) is that the observed data pools random draws from several different underlying distributions.[6] In our case, each underlying distribution that is a component of the mixture represents a different earnings generating

[6]See McLachlan, Lee, and Rathnayake (2019) for an overview of FMMs.

mechanism, where the difference could be one simply of scale or a more fundamental difference. By varying the component types being considered in addition to the number of mixture components, we can allow for heterogeneity both in terms of the number of segments as well as their underlying generating mechanisms.

Since we are looking at fundamentally one-dimensional data (income), such mixture models are relatively straightforward to estimate using Maximum Likelihood Estimation (MLE) and this is typical in the literature.[7] In general we specify the total distribution of income as X where $p[x]$ is a probability density function of X such that $p[x] \geq 0$ and $\int p[x]dx = 1$. Assuming the distribution of income is generated by a K-component finite-mixture distribution, this implies X is a linear combination of K component density functions each with a respective weight λ_k (s.t. $\sum_{k=1}^{K} \lambda_K = 1$), and each parameterized by a vector θ_k. The probability density assigned to making the observation x_i is therefore given by:

$$p[x_i|\theta_1, \ldots, \theta_k] = \sum_{k=1}^{K} \lambda_k \, p_k[x_i|\theta_k] \tag{2}$$

This model defines a likelihood for the N observed incomes $\{x_i\}_N$ conditional on the K-component mixture model:

$$p[\{x_i\}_N|\theta_1, \ldots, \theta_k] = \prod_{i=1}^{N} \sum_{k=1}^{K} \lambda_k \, p_k[x_i|\theta_k] \tag{3}$$

The discovery that a finite mixture model provides a thorough and stable statistical description of the distribution of income shows that heterogeneity inherent in labor markets and allow us to obtain good estimates of the evolution of the parameters for each distribu-

[7] Lubrano and Ndoye (2016) details some of the benefits from Bayesian methods as well.

tional form we identify as well as of their relative weights in the overall distribution. The result is that each labor market segment is endogenously determined by the data itself.

The specific model we fit in this paper is the two-component exponential-log normal mixture model first applied to publicly available CPS data in Schneider (2013). This model takes the following four parameter form:

$$p[x_i|\alpha, \mu, \sigma, \lambda] = \lambda \, \text{Exp}[x_i|\beta] + (1 - \lambda) \, \text{lgN}[x_i|\mu, \sigma] \tag{4}$$

The use of finite mixture models thus allows for overlapping distributions without *ad hoc* sorting of workers into each proposed segment (see Osterman, 1975) with hard cut-off incomes as is common in existing literature (see Dragulescu and Yakovenko, 2000).

To test what type of worker falls into each component of the mixture, we can estimate the component weight, λ, as conditional on worker type. For the initial analysis presented here, we use an administrative dummy variable that codes all workers reporting that they typically work less than 35 hours per week as "part-time" (PT) and all workers that typically work 35 hours or more as "full-time" (FT). The conditional expression of λ is simply:

$$\lambda = \beta_0 + \beta_1 \, FT \tag{5}$$

The estimated propensity for a PT worker to be in the exponential component of the mixture is therefore given by $\hat{\beta}_0$ and that of a FT worker by $\hat{\beta}_0 + \hat{\beta}_1$. If PT workers indeed dominate the exponential component of the mixture, then we expect a negative and statistically significant estimate of β_1.

Model Results

For the sake of brevity, we do not present formal fit comparisons to other models in this paper and instead concentrate on interpreting what the fit of the proposed mixture model given by equation (4) tells us. However, based on Schwarz's Bayesian Information Criteria (BIC), the mixture model provides an overwhelmingly better fit of the earnings data than either the exponential or log-normal components alone. It also provides a better fit than the Gamma distribution, which only very modestly improves on the fit of the exponential. The GB2 distribution does provide a better fit of the data, which we largely attribute to its greater ability to fit a power-law tail.[8] A systematic evaluation of the best model specification is part of our ongoing project.

Figure 2 and 3 plots the evolution of the estimated model parameters (ML estimates are detailed in Appendix A). The parameter β is the inverse of the exponential scale parameter and is the mean of the exponential component of the mixture model. Both μ and β are adjusted for inflation using the CPI.

We now note a few trends in the parameter evolutions and provide some interpretation. The scale parameters of both components of the mixture, when adjusted for inflation, decline over the period 1975 to 2017. This is consistent with stagnant median incomes and slow growth in mean incomes. What our model adds is that mean and median incomes captured in the exponential component declined significantly since 1975 (see the top right panel in fig. 2). If, as we speculate, the exponential component is dominated by workers earning the minimum wage or near it, then this makes sense as the minimum wage persistently eroded in real value from its peak in 1968.

The decline in μ (bottom left panel) is less precipitous and more steady, consistent with the log-normal component reflecting somewhat less precarious employment. The implication

[8]Schneider (2013) using the public-use data and a different fit criteria found that the GB2 and the exponential / lognormal mixture provided comparable fits of the data.

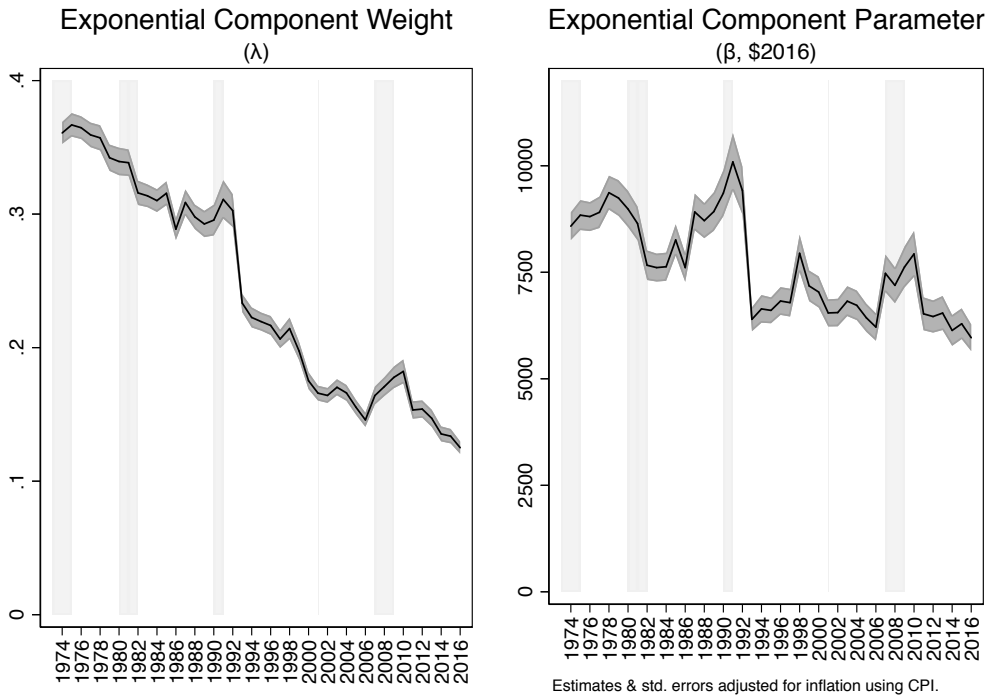


Figure 2: Parameter estimates for the exponential component in the mixture. The shaded region extends two standard errors above and below the line of point estimates.

is still a declining real value of median income for this component. The rise in σ, the shape parameter of the log-normal component, indicates that inequality in that component increase considerably as did the real mean income associated with it. Overall, these parameters therefore indicate a falling behind of the labor market segment captured by the exponential component in addition to growing disparity within the labor market segment captured by the log-normal component.

An unexpected result is that the weight of the exponential component is declining steadily over the entire period. We do not yet have a good explanation for this, although it may reflect that part-time basic labor simply does not pay sufficiently well in real terms. Notable is that the growth in the labor force (estimated from the sum of the weights used in the analysis) offset this decline until around 1990, so that the number of workers associated with the exponential component remained relatively constant from 1975 until 1990. After 1992,

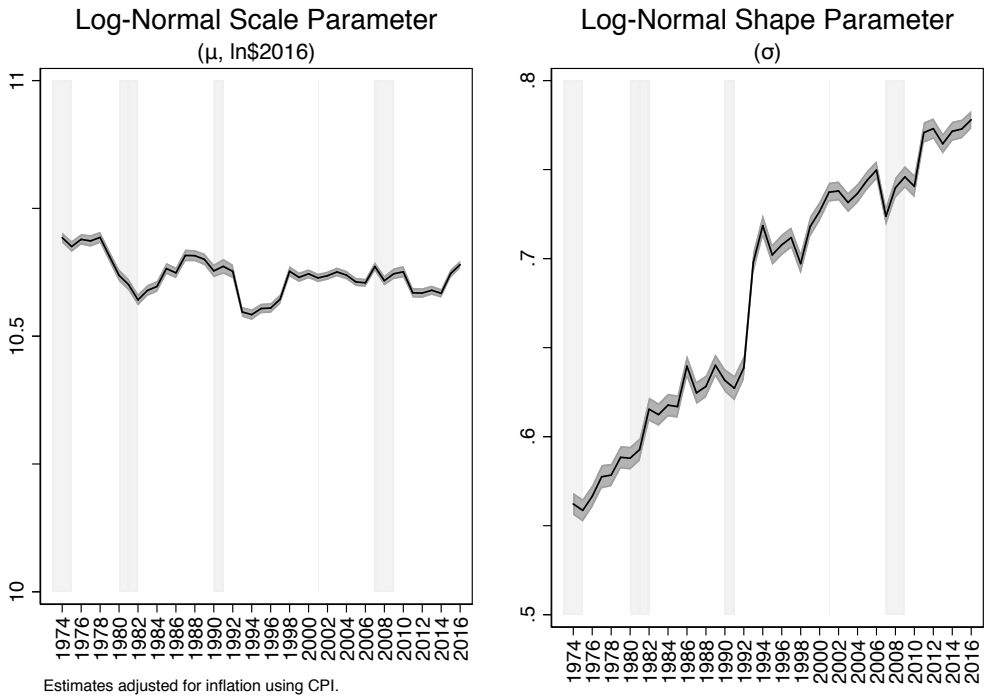


Figure 3: Parameter estimates for the log-normal component in the mixture. The shaded region extends two standard errors above and below the line of point estimates.

the number of workers in the exponential component declined from around 30 million to 20 million because labor force growth slowed. (Note that the trend in λ shown in the top left panel does not show a notable change around 1990-1992.)

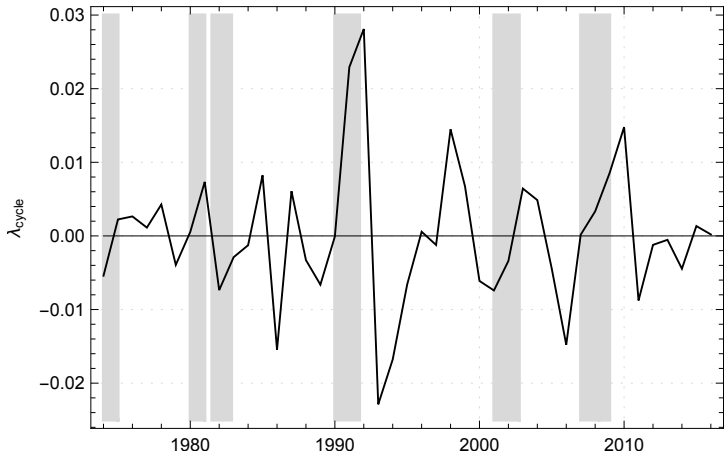


Figure 4: Cyclical variation in HP filtered λ against NBER business cycles.

We want to also note that according to Weitzman (1989)'s interpretation of his model, we should expect the share of more contingent and precarious workers to rise at the onset of a recession as employers switch to production strategies that shifts risk to workers. Since we speculate that such workers are represented by the exponential component of the mixture, this provides a testable hypothesis about the cyclical movement of the exponential component's weight (λ). While there are not enough business cycle downturns in our study window to form any firm conclusions, the prediction does appear consistent with the cyclical movement of λ: each of the downturns in our data are associated with a temporary increase λ.

For further clarity we compare the density evolution of the income distribution as captured by the model in Figure 10 for 1974 and 2016.

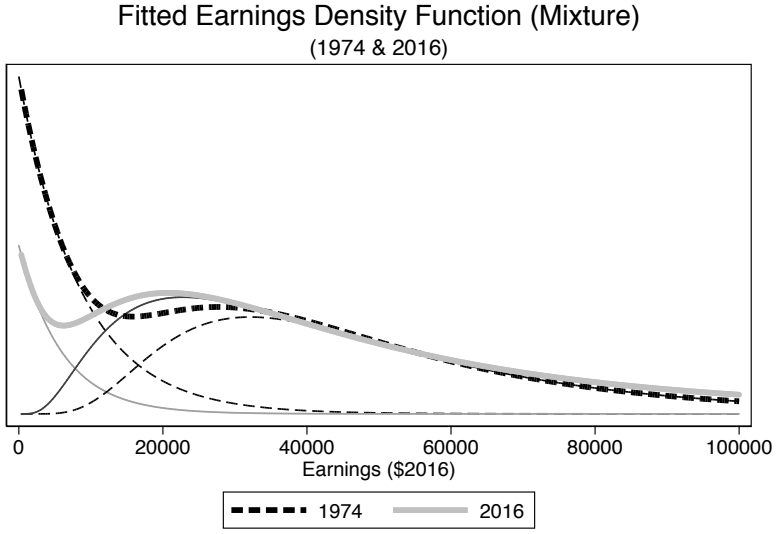


Figure 5: Fitted earnings densities for 1974 and 2016 illustrating the evolution of the distribution of incomes based on the two-component mixture. The thin dashed lines show the components for 1974 (weighted by λ) and the thin solid lines show the components for 2016.

Correlation with Worker Type

As described above, we can re-estimate the model allowing λ to be conditional on worker type (PT or FT); shown in figure 6. The estimated population proportion of FT workers (workers who report working 35 hours or more during a typical week) fluctuates with the business cycle between 82.5% and 86% during the study period around a relatively constant average. The estimate for λ for part-time workers jumps from the values indicated in fig. 2 to 0.97 in the 1970s. Consistent with the trend in fig. 2 (top left panel), the exponential weight for PT workers declines, but only modestly from 0.97 to around 0.93 by the end of the study period. Thus our first finding is that income observations associated with PT workers are overwhelmingly likely to fall in the exponential component.

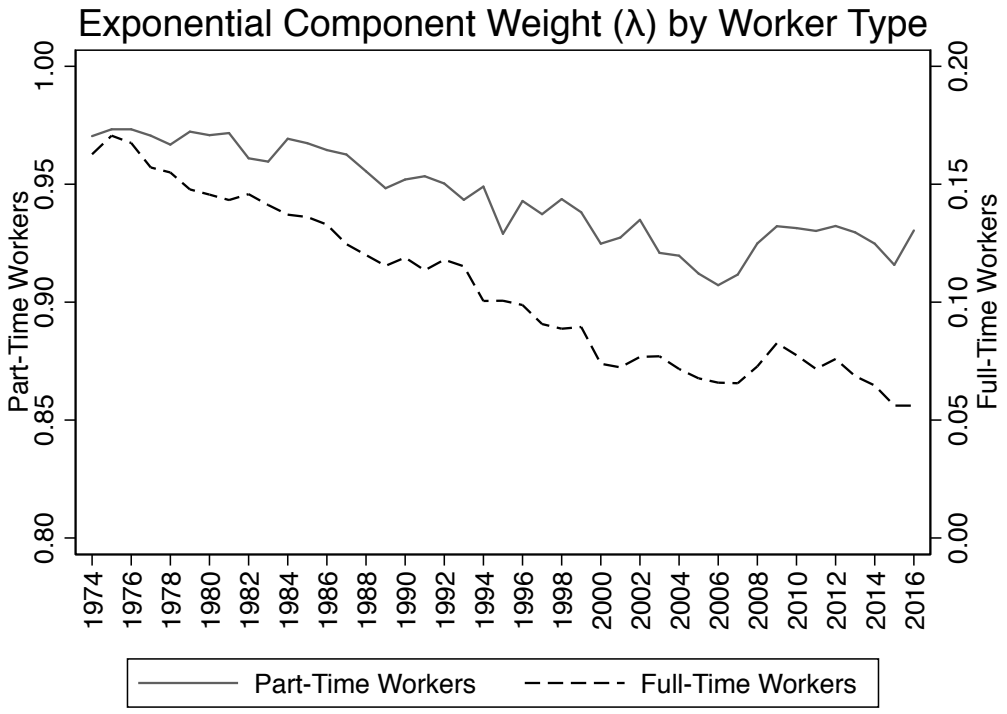


Figure 6: Conditional exponential component weights conditional on worker type (scale for "Full-Time" workers given on right y-axis).

Interpreting the conditional weight of the exponential component as a propensity for a

specific worker type to fall in the exponential component, we can interpret the estimation results as follows. The estimated parameter for β_1 in (5) (the difference in the propensity to be in the exponential component between PT and FT workers) is negative and highly statistically significant for all years. It varies between -0.83 to -0.87, suggesting that the propensity for a FT worker to fall in the exponential component declines from around 11% in the 1970s to just under 6% by 2017 (see the dashed line in fig. 6 read against the right-side y-axis).

The decline in the propensity of full-time workers' earnings to be captured by the exponential component in the mixture is greater in absolute terms and much more dramatic in percentage terms compared to the modest decline in β_0 in (5) over time. Hence, the relative propensity of a full-time worker's earnings to be in the exponential component has decreased significantly over the study period, and this is likely driving the observed decline in λ in fig. 2. The result is striking given that the mode (and mass) of the log-normal component shifted left towards lower incomes between 1974 and 2016 as illustrated in fig. 5, yet the exponential component is almost entirely populated by earnings from individuals identified as part-time workers by 2016. In so far as PT workers are more likely to be younger, people of color, and women, based on these results we would expect to find statistically significant correlations with λ and race, gender, and age in future work. However, some caution is warranted in interpreting these results because the worker type distinction used in this analysis is based on an administrative variable that codes worker's who report working less than 35 hours per week on average as part-time. Specifically workers with multiple part-time jobs that in total work more than this are lumped together with full-time workers, which may distort some of the analysis.

Measures of Inequality

Economists study the distribution of income in order to provide a sound basis for public policy that aims to reduce harmful social and economic effects of inequality. Both earlier empirical work on the dual-labor market (Reich et al., 1973, Osterman, 1975, Weitzman, 1989) and recent work on the dual structure of the US economy (Temin, 2017) suggests that single summary statistics of income inequality are insufficient for understanding the relationship between policy and programs, and their distributional consequences.

The benefit of a segmented labor market approach is not only the ability to identify distinct labor market segments, but also to identify a more detailed measure of inequality in economy by examining the within and between inequalities of each component. Following Dagum (1997) and Anderson et al. (2018) with a population of size N with total income Y partitioned into K (possibly overlapping) subpopulations where n_k represents the population in the kth group such that $\sum_k n_k = N$ and f_k is the distribution function of kth group then, letting the population share and income share in group k be $p_k = \frac{n_k}{N}$ and $s_k = \frac{\mu_k}{\mu}$, where μ_k is the average income in group k, and with G_k representing the weighted Gini of the kth group, the total Gini coefficient can be decomposed into three components:

$$G = G_w + G_b + G_t \tag{6}$$

Where

$$G_w = \sum_{k=1}^{K} p_k^2 s_k G_k \tag{7}$$

measures the contribution of within subpopulation inequality to the total Gini,

$$G_b = \frac{1}{\mu} \sum_{k=2}^{K} \sum_{j=i}^{k-1} p_k p_j |\mu_k - \mu_j| \tag{8}$$

measures the contribution of between subpopulation inequality and

$$G_t = \frac{2}{\mu} \sum_{k=2}^{K} \sum_{j=i}^{k-1} p_k p_j \int_0^{\infty} f_k[y] \int_y^{\infty} f_j[x](x-y)dxdy \tag{9}$$

measures the contribution of the transvariation or overlap between subpopulations. In our model $k = 2$, the population share is $p_1 = \lambda$ and $p_2 = 1 - \lambda$, and $f[x] = \text{lgN}[\mu, \sigma]$ and $f[y] = \text{Exp}[\beta]$. Figure 7 shows the estimated contribution of each component of the Gini decomposition.

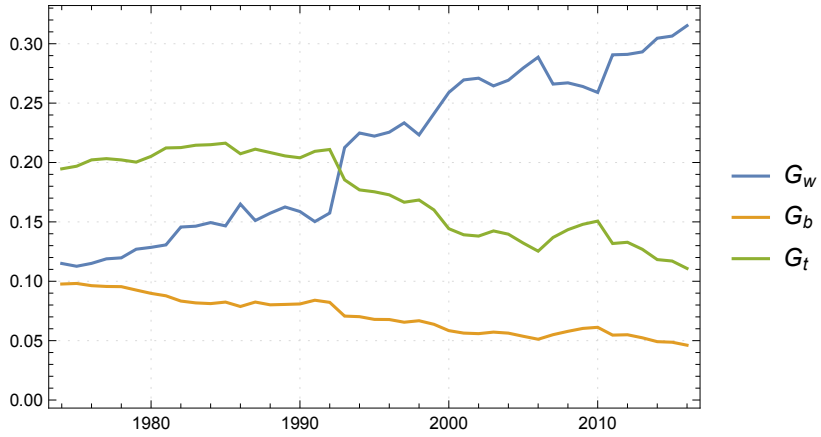


Figure 7: Within, between, and transvariation measures of the decomposed Gini coefficient.

We directly estimate the overall, within, and transvariation component calculate the between inequality as the residual $G_b = G - G_w - G_t$. We can see the driver of the within inequality component by examining the weighted Gini of each segment.

Presumably because we do not accurately capture very high income by not including a power-law component in our mixture and because we focus only on income earned from labor, the overall level of inequality reflected in the fitted model actually declines over the study period. That is at least partially attributable to the decreasing weight of the exponential component. (The exponential distribution has a constant Gini of 0.50, which is higher than the inequality in the log-normal component.) However, inequality within the log-normal component is rising rapidly over the study period.

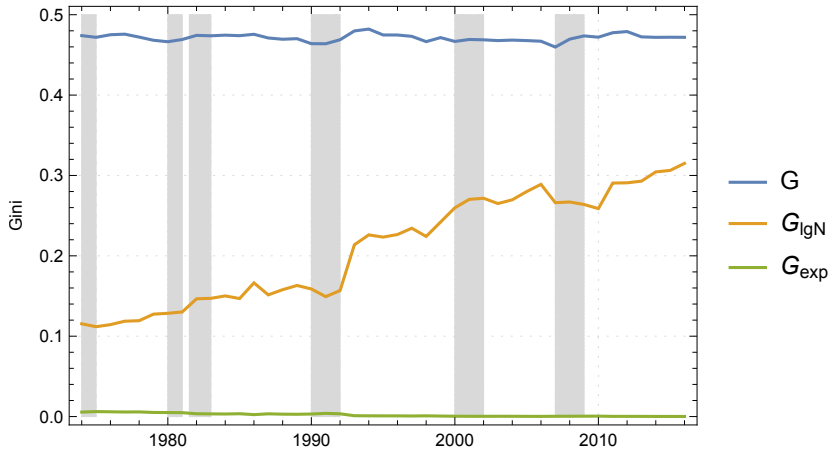


Figure 8: Total inequality and within segment inequality measured by the Gini coefficient.

Because all three components of GINI are non-negative and $0 \leq G_t \leq G$, we can follow Anderson et al. (2018) and define a segmentation index which measures of the degree to which constituent segments do not overlap:

$$\text{SI} = 1 - \frac{G_t}{G} = \frac{\frac{2}{\mu}\sum_{k=2}^{K}\sum_{j=1}^{k-1} p_k p_j \int_0^\infty f_k[y] \int_y^\infty f_j[x](x-y)dxdy}{1 - \frac{1}{\mu}\int_{-\infty}^{\infty}(1-F[x])^2\,dx} \tag{10}$$

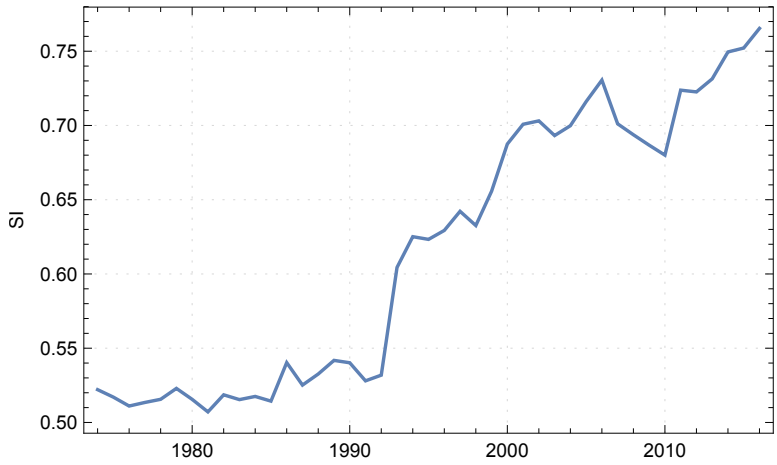


Figure 9: Estimated segmentation index.

Because the transvariation measure in Figure 7 is declining we see that the segmentation index is rising, implying that the upper segment represented by the log-normal component is

being pulled further to the right. This picture appears consistent with the findings of Piketty (2014) who shows the income share of the top earners as increasing dramatically after 1990.

Conclusion

The search for the best functional fit for the observed distribution of income has often ignored fundamental differences in income generating mechanisms across labor markets. The concept of a homogeneous labor market naturally leads to explanations of the variation in income as arising from individual characteristics. This conceptualization of a labor market process operating uniformly on all individuals leads to the modeling of the income distribution in a single encompassing functional form. The theory of labor market segmentation identifies qualitative differences between various labor market processes and has a natural interpretation in finite mixture models. In this paper we have demonstrated the usefulness of finite mixture modeling for analyzing a heterogeneous labor market incorporating the complexities of the various functional forms that arise from theoretical considerations.

Using restricted-access Census data, our initial results indicate that the exponential-log normal mixture model provides a stable fit to the distribution of income for the period 1975-2017. Furthermore, we find that the parameter evolution clarifies details of the changing distribution of income over this period. Most notably, incomes captured by the log-normal component become more dispersed. The exponential component's relative contribution to the mixture appears to shrink while the mean and median income of that component decreases considerably.

Most significantly, we show that the exponential component is associated with part-time workers, supporting our prior that it represents the labor market for workers supplying basic labor and facing highly variable hours of work.

Decomposing the Gini coefficient, we also demonstrate that labor markets are becoming more segmented as indicated by the rising segmentation index. This segmentation appears

to be worsening considerable since 1990 in line with the rising income share of top income earners.

Appendix A: Data

Data is made available through the Rocky Mountain Federal Statistical Research Data Center (RDC) project #1935. Summary statistics from the model are presented in following table.

YEAR	N_OBS	POP_EST	A	SE_A	BETA	SE_BETA	MU	SE_MU	SIG	SE_SIG
1974	62000	97400000	0.36	0.00	1763	33.40	9.11	0.01	0.56	0.00
1975	60500	96800000	0.37	0.00	1982	39.30	9.18	0.01	0.56	0.00
1976	73500	99400000	0.36	0.00	2088	40.30	9.25	0.01	0.57	0.00
1977	72500	101900000	0.36	0.00	2249	46.60	9.31	0.01	0.58	0.00
1978	73500	105000000	0.36	0.00	2545	54.00	9.39	0.01	0.58	0.00
1979	87000	108800000	0.34	0.00	2796	63.40	9.46	0.01	0.59	0.00
1980	87000	109500000	0.34	0.01	3084	73.00	9.55	0.01	0.59	0.00
1981	77500	110200000	0.34	0.00	3274	75.20	9.63	0.01	0.59	0.00
1982	76000	109400000	0.32	0.00	3082	69.70	9.66	0.01	0.62	0.00
1983	76000	110800000	0.31	0.00	3158	68.10	9.71	0.01	0.61	0.00
1984	77500	114400000	0.31	0.00	3303	71.60	9.76	0.01	0.62	0.00
1985	76500	117000000	0.32	0.00	3703	81.90	9.83	0.01	0.62	0.00
1986	76000	118800000	0.29	0.00	3475	76.20	9.84	0.01	0.64	0.00
1987	76500	120800000	0.31	0.00	4219	101.70	9.91	0.01	0.62	0.00
1988	72000	123000000	0.30	0.00	4294	101.40	9.95	0.01	0.63	0.00
1989	78500	124400000	0.29	0.00	4611	116.80	9.99	0.01	0.64	0.00
1990	77500	124600000	0.30	0.01	5095	147.10	10.02	0.01	0.63	0.00
1991	76000	124700000	0.31	0.01	5727	191.60	10.07	0.01	0.63	0.00
1992	75000	126100000	0.30	0.01	5507	164.10	10.09	0.01	0.64	0.00
1993	72500	127400000	0.23	0.00	3851	84.20	10.04	0.00	0.70	0.00
1994	72500	129900000	0.22	0.00	4100	100.30	10.06	0.01	0.72	0.00
1995	64500	132900000	0.22	0.00	4195	96.50	10.10	0.00	0.70	0.00
1996	65500	135500000	0.22	0.00	4462	106.50	10.13	0.00	0.71	0.00
1997	65000	135400000	0.21	0.00	4540	109.10	10.17	0.00	0.71	0.00
1998	65500	137200000	0.21	0.00	5396	148.20	10.24	0.01	0.70	0.00
1999	67500	140100000	0.20	0.00	4984	124.80	10.25	0.00	0.72	0.00
2000	65000	141000000	0.18	0.00	5049	131.40	10.29	0.00	0.73	0.00
2001	107000	143000000	0.17	0.00	4830	117.70	10.31	0.00	0.74	0.00
2002	105000	143000000	0.16	0.00	4913	121.40	10.33	0.00	0.74	0.00
2003	103000	142800000	0.17	0.00	5230	134.30	10.36	0.00	0.73	0.00
2004	101000	143900000	0.17	0.00	5292	136.70	10.38	0.00	0.74	0.00
2005	101000	145900000	0.16	0.00	5235	134.50	10.40	0.00	0.74	0.00
2006	100000	148000000	0.15	0.00	5220	134.80	10.43	0.00	0.75	0.00
2007	101000	149400000	0.16	0.00	6458	189.90	10.49	0.00	0.72	0.00
2008	101000	149300000	0.17	0.00	6452	187.80	10.50	0.00	0.74	0.00
2009	98500	145700000	0.18	0.00	6814	214.80	10.51	0.01	0.75	0.00
2010	95500	144300000	0.18	0.00	7205	244.20	10.53	0.01	0.74	0.00
2011	93500	145700000	0.15	0.00	6115	180.90	10.52	0.00	0.77	0.00
2012	94500	148300000	0.15	0.00	6181	180.50	10.54	0.00	0.77	0.00
2013	93500	149800000	0.15	0.00	6351	193.80	10.56	0.00	0.76	0.00
2014	93000	151200000	0.14	0.00	6052	178.30	10.57	0.00	0.77	0.00
2015	87000	154100000	0.13	0.00	6213	177.90	10.61	0.00	0.77	0.00
2016	87500	155700000	0.13	0.00	5964	156.20	10.64	0.00	0.78	0.00

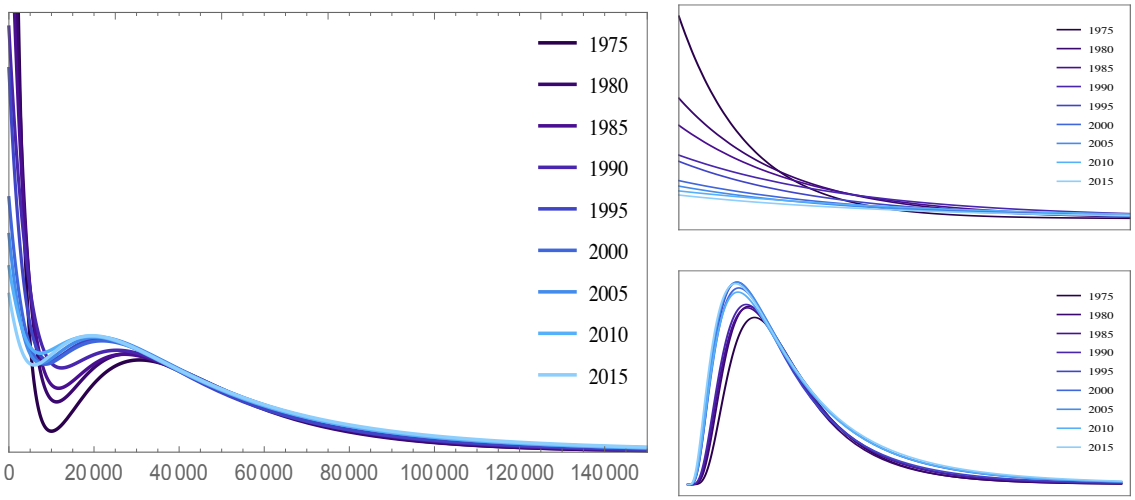


Figure 10: Evolution of mixture model and individual components

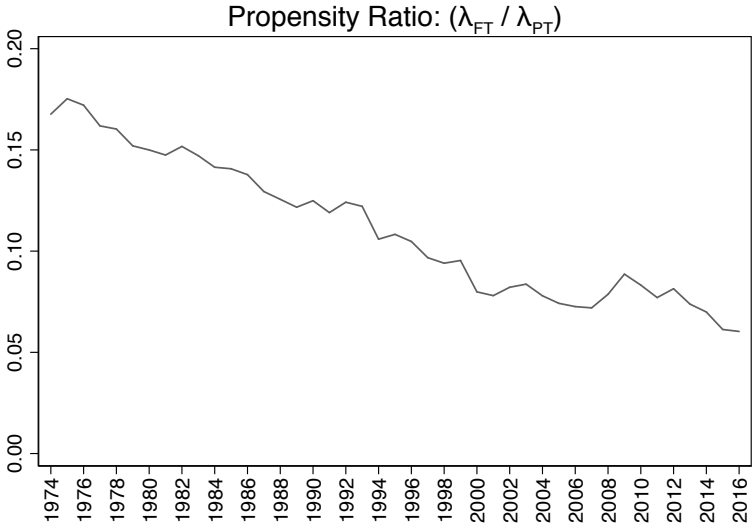


Figure 11: Relative propensity for a full-time worker to be part of the exponential component of the mixture.

Appendix B: Price-adjusted Parameters

The adjustment from nominal to real values to which the mixture model boils down to multiplying the nominal values by some price-adjustment factor, c. (In our adjustments, c is the ratio of CPI for the base year to current year, but any other index would obviously imply the same operation.) The question thus is simply how the parameters are affected

when we transform a random variable X to $Y = c\,X$, which turns out to be straightforward. If the *pdf* of X is $p_X[x, \theta]$, then

$$p_Y[y, c, \theta] = \frac{1}{c} \cdot p_X[y/c, \theta]$$

In the specific case of the exponential distribution, where $p_X[x, \beta] = \frac{1}{\beta} \exp\left[-\frac{x}{\beta}\right]$, this transformation implies that $p_Y[y, c, \beta] = \frac{1}{c\,\beta} \exp\left[-\frac{y}{c\,\beta}\right]$. Or alternatively, if $X \sim \text{Exp}[\beta]$ then $Y \sim \text{Exp}[c\,\beta]$. Similarly, it is trivial to work out that if $p_X[x, \lambda, \beta, \mu, \sigma] = \lambda\,\text{Exp}[\beta] + (1 - \lambda)\,\text{lgN}[\mu, \sigma]$, then

$$p_Y[y, c, \lambda, \beta, \mu, \sigma] = \lambda\,\text{Exp}[c\,\beta] + (1 - \lambda)\,\text{lgN}[\mu + \ln c, \sigma]$$

In other words, the only implication of transforming the underlying values from nominal to real is that the scale parameters should adjust so that $\beta \to c\,\beta$ and $\mu \to \mu + \ln c$.

Since we consider alternative fits in the course of this research, it is worth noting that the same applies to other distributions commonly used in this context. For example, the Gamma distribution ($X \sim \Gamma[a, b]$ then $Y \sim \Gamma[a, c\,b]$), the Dagum distribution ($X \sim \text{Dag}[p, a, b]$ then $Y \sim \text{Dag}[p, a, c\,b]$), and the generalized Beta distribution of the second kind (or GB2; $X \sim \text{GB2}[p, q, a, b]$ then $Y \sim \text{GB2}[p, q, a, c\,b]$). Notably, for each of these the affected parameter does not affect the kurtosis of the distribution, so the fatness of the tail is unaffected by the scale translation of changing underlying values from nominal to real.

Appendix C: Simple Earnings Models

Here we briefly outline the core argument for how a log-normal and exponential distribution of earnings might arise. First, we focus on the log-normal and Gibrat's Law. However, our focus is the position that an individual occupies in a given firm, not the individual who occupies it at any given moment. The reason is twofold: the position is likely longer-lived

than any individual's tenure at the firm (possible longer than any individual's lifetime) and we would expect strong correlation of remuneration for any individual(s) occupying that position from one period to the next. (We are purposefully non-committal as to whether there is substitutability between positions as there might be in a neoclassical setup or if positions appear in fixed proportions within a given firm.)

The point is that a given firm requires a certain set of positions to be filled and pay associated with each is determined by the role of that position within the firm's production process. Initially, that pay is y_0. From one period to the next, the pay to each position is adjusted and we let the adjustment be some random percentage increase, ϵ_t, from the original pay. So period 1 pay is:

$$y_1 = (1 - \epsilon_t)\, y_0$$

and after T periods, $y_T = \prod_{t=1}^{T} (1 - \epsilon_t)\, y_0$. The heroic assumption on our part is that each ϵ_t is i.i.d. with mean m and variance $\sigma^2 < \infty$. To see how heroic this assumption is consider everything that is being included in the stochastic raise: cost-of-living adjustments, specific pay adjustments for hiring an over- or underqualified person to fill the position (i.e. the individual characteristics of the individual filling the position and thus changes in the individual filling the position), changes in technology that result in a revaluation of the relative contribution of the position to production, etc. Lumping these things together and considering them random while also postulating that across positions they are i.i.d. makes ϵ_t a very comprehensive catch-all.

Given that assumption and furthermore assuming that m is small (much closer to zero than to unity; practically less than 0.1), we can make the usual simplification by taking the log of the above expression and then approximating $\ln(1 + \epsilon_t) \approx \epsilon_t$.

$$\ln y_T = \sum_{t=1}^{T} t \, \ln(1 - \epsilon_t) + \ln y_0 \approx \sum_{t=1}^{T} \epsilon_t + \ln y_0$$

Applying the Central Limit Theorem, we can make the case that $\ln y_T$ will be approximately normally distributed. Furthermore, the initial value y_0 will play a vanishingly small role in determining the mean of the distribution as long as $T \cdot m >> y_0$. In other words, far from the inception period, the distribution of pay associated with different positions is more determined by the stochastic evolution over many periods than their original pay based on their relative contribution to production. Here it is important, however, to remember that the stochastic evolution itself included technological changes that rearrange the relative value of different position within the production process. To put it concretely, a position that started as a highly paid job might well end up towards the lower end of the pay spectrum if technological change has made the particular skillset close to obsolete over the years. We are treating that evolution as part of the "random" shocks ϵ_t. The end result is that y_T is asymptotically log-normally distributed and this distribution holds for a number of different positions across the economy as long as the associated specific starting salaries associated with them are small compared to the accumulated effects of cost-of-living pay adjustments, technological change, etc.

Maintaining a similar focus on positions at firms that are longer lived than individuals (or anyway their tenure in a specific position), we now turn to the exponential distribution. We posit that there is some work that can be done by any employee and efficacy with which it is done is the same across employees. The only thing that matters for these tasks is that the work gets done, not by whom or their particular skill set. In very short time frames, the total work of this nature is probably a fixed amount at the firm level. So maybe we can imagine routinized service work that fundamentally amounts to little more than having an employee behind the counter during open-for-business hours.

Formally, it may be very difficult to assign remuneration to this type of work because it has a discrete impact on firm operations. If it does not get done, none of the other productive activity succeed and its marginal contribution is infinite; if it gets done, production succeeds and any more of it leads to no more production so that it's marginal contribution is zero. Some firms may solve this problem by enticing or coercing existing employees in positions not dedicated to such tasks to complete such them in addition to their direct responsibilities essentially for free (or very low pay). Other firms, particularly in the service sector, may hire part-time employees who are essentially interchangeable and with overall sufficiently flexible schedules that these basic tasks can always be covered. If one employee is not available, another gets their hours that week, for example.

In very short time frames, the total amount of time required for these tasks across employees is fixed and furthermore changes between short periods appear like a binary exchange in hours between workers. Finally, if the workers filling these kinds of positions are paid approximately the same (which would almost certainly be the case within each establishment, but is likely to even hold across establishments), then this all amounts to an argument how an exponential distribution of earnings among these workers might arise.

Formally, the argument from physics that applies here is that the number of hours any two employees (among many) supply is determined by their respective probabilities of agreeing to work a certain number of hours, x_1 and x_2 respectively. We assume that the probability functions are the same for each employee, so that the probability of them together working $x_1 + x_2$ hours is $f[x_1] \cdot f[x_2]$. On the other hand, the total amount of time required is fixed, X, and thus the particular combination of $x_1 + x_2$ implies that the overall scheduling arrangement must be such that $X - (x_1 + x_2)$. If all scheduling arrangements are considered equally likely, then the probability of such an overall scheduling arrangement must be equal to the probability of $x_1 + x_2$. Thus, the probability distribution function we are looking for has the property that:

$$f[x_1] \cdot f[x_2] = h(x_1 + x_2)$$

The only function that will satisfy this equality is the exponential, thus suggesting that the choice of hours across employees will have follow an exponential distribution. Furthermore, if all such employees are paid the same wage, then their earnings distribution will also be exponential. The key to making this line of reasoning plausible is that the required schedule fluctuations amongst employees happen at a much faster time scale than changes in the aggregate hours of basic routine work across business. As we have presented it, we envision schedule changes week-to-week and suspect that total hours required change no faster than quarterly and maybe only annually. Thus the "churn" in schedules is 10 to 50 times faster than growth in the aggregate constraint.

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